Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/02/2015
PI Name:	Davis, Catherine M. Ph.D.		
Project Title:	Neurobehavioral and CNS-Related Physiological Changes Following Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	catherine.davis-takacs@usuhs.edu	**Fax:**	FY 301-400-4023
PI Organization Type:	NON-PROFIT	**Phone:**	301-400-4596
Organization Name:	Henry M. Jackson Foundation		
PI Address 1:	6720-A Rockledge Dr.		
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City:	Bethesda	**State:**	MD
Zip Code:	20817-1891	**Congressional District:**	8
Comments:	Campus address (Jan 2022): Department of Pharmacology and Molecular Therapeutics, Uniformed Services University of the Health Sciences, 4301 Jones Bridge Road, Bethesda, MD 20814. NOTE: PI formerly at Johns Hopkins University; moved to Henry M. Jackson Foundation for the Advancement of Military Medicine in fall 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	01/01/2015	**End Date:**	12/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert Ph.D. (Johns Hopkins University)		
Grant/Contract No.:	NNX15AC71G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Assessing the biological consequences of living in the space radiation environment represents one of the highest priority areas of NASA research. Of critical importance is the need for an assessment of the vulnerabilities of the central nervous system (CNS) leading to functional neurobehavioral changes during long-term space missions, and the development of effective countermeasures to such risks. The present proposal addresses this need via the application of an animal model to: 1) monitor indices of circadian function (i.e., spontaneous locomotor activity, body temperature) over various time points following irradiation; 2) determine the degree to which changes in circadian function are correlated with the severity of behavioral deficits in a rat psychomotor vigilance task; and 3) examine the underlying neurochemical alterations that accompany these radiation-induced changes in circadian function and neurobehavioral performance. Prior research has 1) identified rats that are sensitive to radiation-induced deficits in neurobehavioral function; 2) identified changes in the dopamine neurotransmitter systems of these radiation-sensitive rats; and 3) identified an increased inflammatory response in the brain from 1-14 days post-irradiation. The current proposal will determine how the immediate effects of irradiation impact subsequent neurobehavioral deficits by assessing physiological markers of circadian function that are sensitive to changes in immunological functions and the onset of pathophysiological states. Groups of animals with implanted biotelemetry transmitters will be trained on a rodent version of the human Psychomotor Vigilance Test (the rPVT) and exposed to radiation; fluctuations in spontaneous locomotor activity and body temperatures will be assessed immediately following exposure and tracked over time and compared to the neurobehavioral performances. Individual variations in these physiological responses following radiation will be assessed for use as possible biomarkers for radiation sensitivity resulting in neurobehavioral deficits. Likely mechanisms of damage to the CNS following radiation exposure will be examined using brain tissue, in addition to an examination of cytokine levels in blood and peripheral organ systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this work will provide knowledge regarding the effects of radiation on neurobehavioral function and circadian fluctuations in core body temperature and locomotor activity. This information will enhance our understanding of radiation's effects and can be used to determine the effects of terrestrial radiation exposures and ways to mitigate their adverse effects.
Task Progress:	In the first year of this grant, the telemetry system was acquired and set up in the lab. A pilot study of 17 rats was conducted in which the rats were trained on the rodent psychomotor vigilance test (rPVT), implanted with biotelemetry devices to record core body temperature and spontaneous locomotor activity, and exposed to x-ray radiation of the frontal cortex only, using the small animal radiation research platform in the Department of Radiation Oncology and Molecular Radiation Sciences. Thus far, rats with x-ray irradiation of the frontal cortex display an increase in median reaction time; i.e., exposed animals respond more slowly to the visual target, while still maintaining normal pre-irradiation accuracy (i.e., percent correct responses). The telemetry data is currently being analyzed. Additionally, 40 rats have been trained on the rPVT, implanted with biotelemetry devices, and will be shipped to Brookhaven National Laboratory (BNL) in November 2015. These rats will be exposed to whole-body protons (150 MeV/n) at 25 and 100 cGy. Telemetry data will be recorded at BNL prior to the radiation exposure and for the days after the exposure, until the rats return to Johns Hopkins. Once back in the lab, telemetry data will be acquired 24-hrs/day and rats will perform their behavioral test once a day, five days per week.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Gupta S, Davis CM, Hienz RD. "Differences in quinpirole-induced yawning between radiation-sensitive and insensitive rats." Presented at Experimental Biology 2015, Boston, MA, March 28-April 1, 2015. FASEB Journal. 2015 Apr;29(1 Suppl):769.3. , Apr-2015
Articles in Peer-reviewed Journals	Davis CM, DeCicco-Skinner KL, Hienz RD. "Deficits in sustained attention and changes in dopaminergic protein levels following exposure to proton radiation are related to basal dopaminergic function." PLoS ONE, in press, as of November 2015.. , Nov-2015
Articles in Peer-reviewed Journals	Davis CM, Roma PG, Hienz RD. "A rodent model of the human psychomotor vigilance test: Performance comparisons." J Neurosci Meth, in press, as of November 2015. , Nov-2015
Awards	Gupta S, Davis CM, Hienz RD. "1st Place for ASPET's Behavioral Pharmacology Division Undergraduate Best Abstract Competition for: Gupta S, Davis CM, Hienz RD. Differences in Quinpirole-Induced Yawning between Radiation Sensitive and Insensitive Rats. American Society for Pharmacology and Experimental Therapeutics, 2015 Experimental Biology Meeting, Boston, MA, April 2015." Apr-2015
Significant Media Coverage	Davis CM, Gupta S. "Interview for the importance of undergraduates attending national meetings and presenting data." Everyday Biochemistry: Official Meeting Blog, https://everydaybiochemistry.wordpress.com/2015/04/02/why-undergraduates-should-attend-eb/ ; accessed 1/3/15., Apr-2015